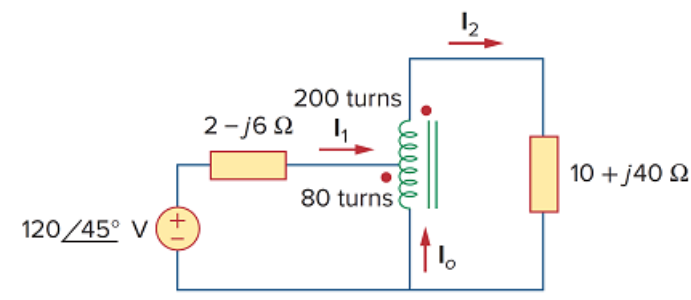


Problem

In the ideal autotransformer of Fig. 13.131, calculate I_1 , I_2 , and I_o . Find the average power delivered to the load.

Figure 13.131



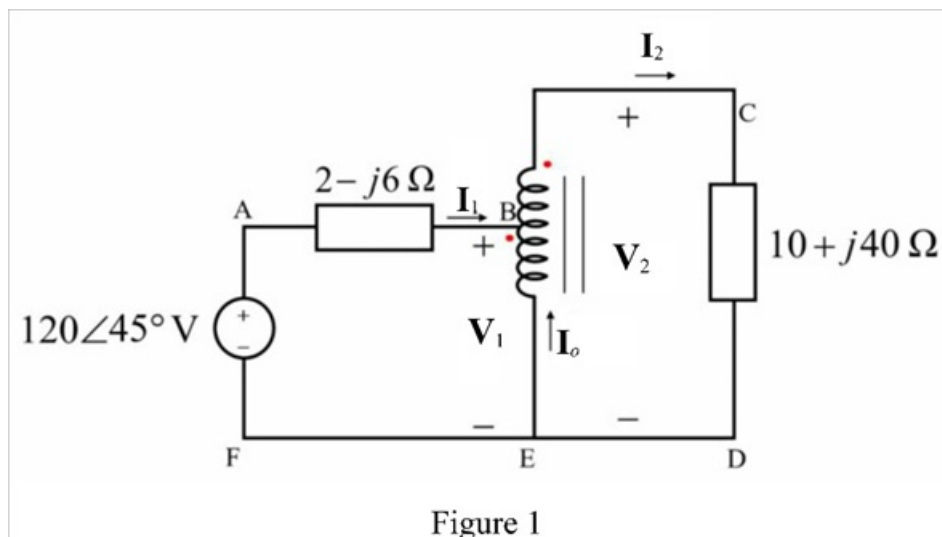
Step-by-step solution

Step 1 of 7

Refer to Figure 13.131 from the textbook.

Observe that the circuit is a step-up auto transformer.

Redraw the circuit as shown in Figure 1.



Step 2 of 7

Apply the Kirchhoff's Voltage Law in the loop ABEF.

$$120\angle 45^\circ = (2 - j6)I_1 + V_1 \dots\dots (1)$$

Apply the Kirchhoff's Voltage Law in the loop BCDE.

$$V_2 = (10 + j40)I_2 \dots\dots (2)$$

Step 3 of 7

For an autotransformer,

$$\frac{V_1}{V_2} = \frac{N_1}{N_1 + N_2} \dots\dots (3)$$

and

$$\frac{I_2}{I_1} = \frac{N_1}{N_1 + N_2} \dots\dots (4)$$

Here, V_1 is the primary side voltage, V_2 is the secondary side voltage, N_1 is the number of primary turns, and N_2 is the number of secondary turns.

Step 4 of 7

Substitute 200 for N_1 and 80 for N_2 in equation (3).

$$\frac{V_1}{V_2} = \frac{200}{200 + 80}$$

$$\frac{V_1}{V_2} = \frac{200}{280}$$

$$\frac{V_1}{V_2} = \frac{5}{7}$$

$$V_2 = \frac{7}{5} V_1 \dots\dots (5)$$

Substitute 200 for N_1 and 80 for N_2 in equation (4).

$$\frac{I_2}{I_1} = \frac{200}{200 + 80}$$

$$\frac{I_2}{I_1} = \frac{200}{280}$$

$$\frac{I_2}{I_1} = \frac{5}{7}$$

$$I_2 = \frac{5}{7} I_1 \dots\dots (6)$$

Step 5 of 7

Substitute $\frac{5}{7}\mathbf{I}_1$ for $\mathbf{I}_2$ and $\frac{7}{5}\mathbf{V}_1$ for $\mathbf{V}_2$ in equation (2).

$$\frac{7}{5}\mathbf{V}_1 = (10 + j40)\frac{5}{7}\mathbf{I}_1$$

$$\mathbf{V}_1 = (10 + j40)\left(\frac{25}{49}\right)\mathbf{I}_1$$

Substitute $(10 + j40)\left(\frac{25}{49}\right)\mathbf{I}_1$ for $\mathbf{V}_1$ in equation (1).

$$120\angle 45^\circ = (2 - j6)\mathbf{I}_1 + (10 + j40)\left(\frac{25}{49}\right)\mathbf{I}_1$$

$$120\angle 45^\circ = \left(2 - j6 + \frac{250 + j1000}{49}\right)\mathbf{I}_1$$

$$120\angle 45^\circ = \left(\frac{348 + j706}{49}\right)\mathbf{I}_1$$

$$120\angle 45^\circ = (7.102 + j14.408)\mathbf{I}_1$$

$$\mathbf{I}_1 = \frac{120\angle 45^\circ}{(7.102 + j14.408)}$$

$$= \frac{120\angle 45^\circ}{16.063\angle 63.76^\circ}$$

$$= 7.47\angle -18.76^\circ \text{ A}$$

Step 6 of 7

Substitute $7.47\angle -18.76^\circ \text{ A}$ for $\mathbf{I}_1$ in equation (6).

$$\mathbf{I}_2 = \left(\frac{5}{7}\right)(7.47\angle -18.76^\circ)$$

$$= 5.336\angle -18.76^\circ \text{ A}$$

Calculate the current $\mathbf{I}_o$.

$$\mathbf{I}_o = \mathbf{I}_1 - \mathbf{I}_2$$

Substitute $\frac{5}{7}\mathbf{I}_1$ for $\mathbf{I}_2$.

$$\mathbf{I}_o = \mathbf{I}_1 - \frac{5}{7}\mathbf{I}_1$$

$$= \left(\frac{7-5}{7}\right)\mathbf{I}_1$$

$$= \frac{2}{7}\mathbf{I}_1$$

Substitute $7.47\angle -18.76^\circ \text{ A}$ for $\mathbf{I}_1$.

$$\mathbf{I}_o = \frac{2}{7}(7.47\angle -18.76^\circ)$$

$$= 2.13\angle -18.76^\circ \text{ A}$$

Step 7 of 7

The load impedance is $\mathbf{Z} = 10 + j40 \, \Omega$. To calculate the average power, consider only the real part of $\mathbf{Z}$, that is, $\mathbf{Z}_{real} = 10 \, \Omega$.

Write the expression for the power delivered (P) to the load.

$$P = |\mathbf{I}_2|^2 \mathbf{Z}_{real}$$

Substitute $5.336 \, \text{A}$ for $|\mathbf{I}_2|$ and $10 \, \Omega$ for $\mathbf{Z}_{real}$.

$$\begin{aligned} P &= (5.336)^2 (10) \\ &= (28.47)(10) \\ &= 284.7 \, \text{W} \end{aligned}$$

Hence, the currents $\mathbf{I}_1$ is $7.47 \angle -18.76^\circ \, \text{A}$, $\mathbf{I}_2$ is $5.336 \angle -18.76^\circ \, \text{A}$, and $\mathbf{I}_o$ is

$2.13 \angle -18.76^\circ \, \text{A}$, and the power delivered to the load (P) is $284.7 \, \text{W}$.